Lecture-2 Sep-24.

Interfaces and Cables.

RJ-45 (RJ= Registered Jack).

- **Ethernet**

  Ethernet is a collection of network protocols/standards.

- **Network Protocols.**

  If there are 2 people, one speaks only english & other speaks only Japanese, there's not going to be any communication between them.

  So, what they need is agreed upon system of communicating like a common language between them.

  Network protocols serve that purpose for network devices.

- **Bits and bytes.**

  Connections between devices in a network operate at a set speed. These speeds are measured in bits per second.

  Bit is a value represented by either 0 or 1.

  8 bits = 1 byte { Eg: 0 0 1 0 1 1 0 1

- Speed is measured in bits per second (Kbps, Gbps, etc)

  Gigabyte = 8 times larger than gigabit
  = 8x Gigabit.

  - 1 kilobit = 1,000 bits
  - 1 megabit (Mb) = 1,000,000 bits (1 million bits
  - 1 gigabit (Gb) = 1,000,000,000 bits (1 billion bits)
  - 1 terabit (Tb) = 1,000,000,000,000 (1 trillion bits)

- Ethernet Standards.
  - Defined in the IEEE 802·3 standard in 1983.
  - IEEE = Institute of Electrical and Electronics Engineers.

Ethernet Standards

Speed	Common Name	IEEE Standard	Informal Name	Maximum Length
10 Mbps	Ethernet	802·3j	10 BASE-T	100m
100 Mbps	Fast Ethernet	802.3u	100 BASE-T	100m
1 Gbps	Gigabit Ethernet	802·3ab	1000 BASE-T	100m
10 Gbps	10 GiG Ethernet	802·3an	10 GBASE-T	100m.

In informal name there is Base & T.

~~BASE~~ BASE = refers to baseband signalling

T = twisted pair cabling.

Max length - That's the maximum length for twisted pair cables as defined in Ethernet for performance & technical reasons.

- UTP cables. (Unshielded Twisted Pair)

The copper cables used in Ethernet standards are UTP cables. Unshielded means that the wires have no metallic shield, which can make them vulnerable to to electrical interference.

↳ #: The twist actually helps protect against electromagnetic interference, or EMI.

There are four pairs of wires twisted together ie 8 wires in total. But not all of the ethernet standards we saw actually use all 8 wires.

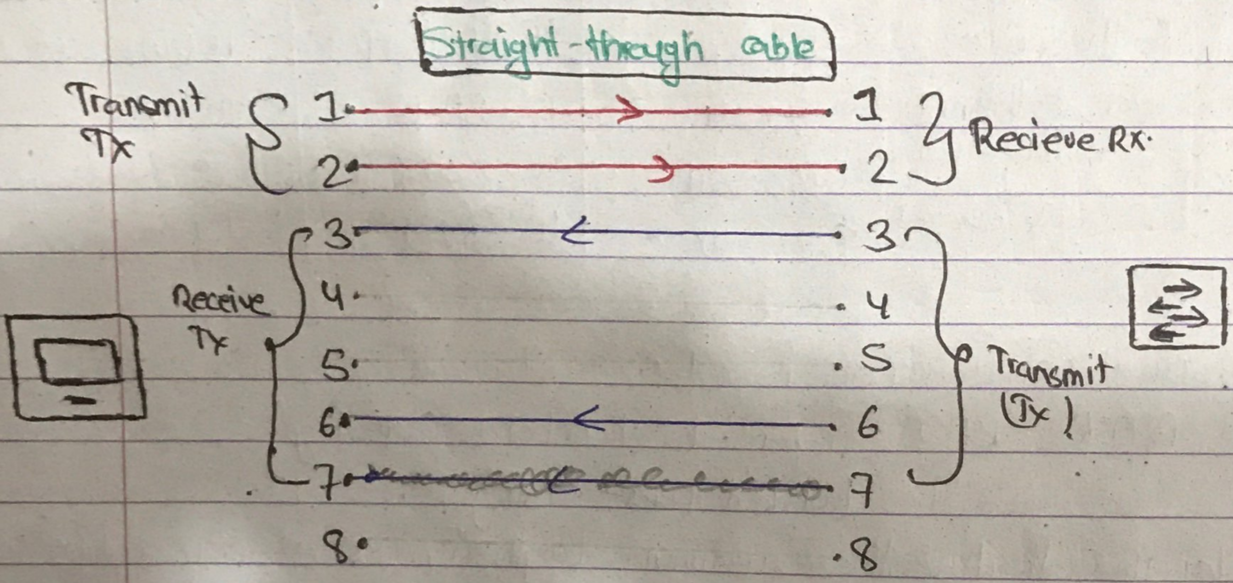
10 BASE-T = 2 pairs (4 wires)

100 BASE-T = 2 pairs (4 wires)

1000 BASET = 4 pairs (8 wires).
10G BASE-T = 4 pairs

Let's see 10-BASE-T and 100 BASE-T

Suppose we connected our PC to switch using 10-BaseT or 100 BASE-T ethernet.



PC sends data throug pin 1 & 2, switch receive from 1&2.
PC receive from 3,6 and switch sends from 3,6.

This allows for full duplex transmission.

Full duplex transmission

Full duplex means that both devices can send and receive data at same time and no problems like collisions will occur because they use separate wires to transmit and receive data.

Same as above will happen with switch & router. Router does same as PC, it sends data from 1&2 and receive from 3 and 6.

A copper Ethernet cable has two RJ-45 connnectors, one on ~~each~~ each end.

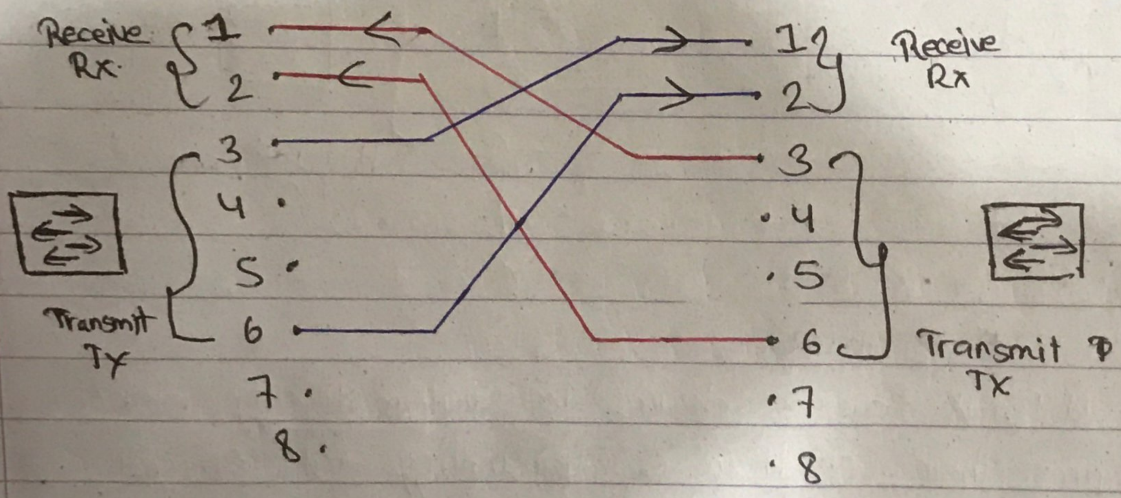
Switch receives on 1,2 & sends from 3 &7

यो router, switch र switch, PC मा use आउने cable हरुलाई straight through cable भनिन्छ,

This kind of cable is called a 'straight through' cable because pin 1 on one end connects a straight through pin 1 on other end.

Lets see connection of two switch using 10 Base-T, 100Base-T.

Crossover Cable



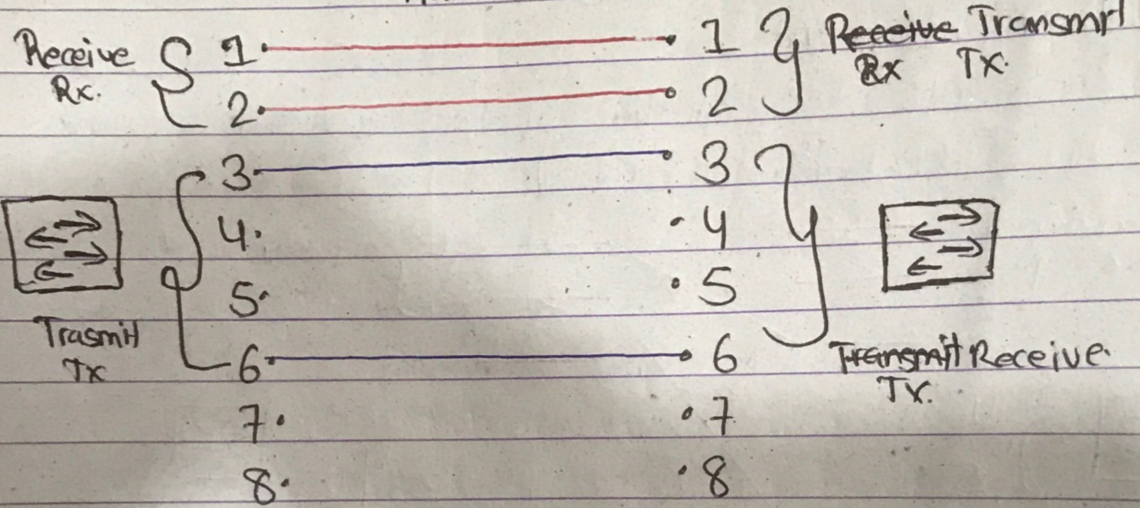
Switch, Switch or PC, router मा we use crossover cable because PC, router मा दुवै को Pin 1,2 ले transmit गर्छ so St.cable can't be used.

Device Type	Transmit (Tx) Pins	Receive (Rx) Pins
Router	1 and 2	3 and 6
Firewall	1 and 2	3 and 6
PC	1 and 2	3 and 6
Switch	3 and 6	1 and 2.

(old device मा mostly used).

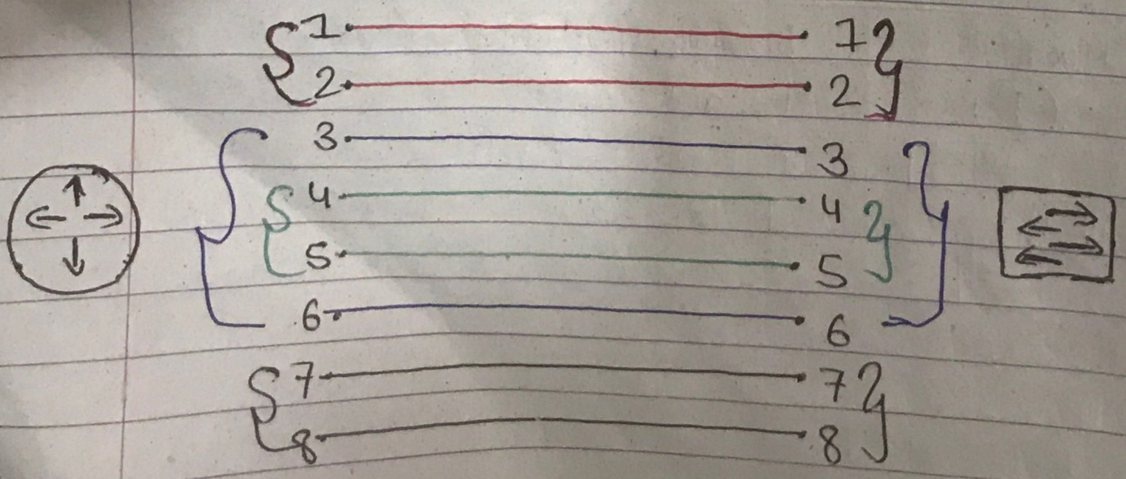
Tyo crossover cable चाँहि पहिला मात्र use गर्नुपर्थ्यो आजभोली भने there's a feature called AUTO MDI-X on newer ~~de~~ networking devices.

Auto MDI-X



Auto MDI-X allows devices to automatically detect which pins their neighbour is transmitting data on, and adjust which pins they use to transmit and receive data.

UTP cables (1000 BASE-T, 10G BASE-T).



1000 Base-T and 10GBASE-T - uses all four pairs (8) of wire; each pair is bidirectional meaning each pair isn't dedicated specifically to transmitting data or receiving data.

Fiber optics Connection.

SFP (Small Form-Factor Pluggable) Transceiver.

Fiber optics is a newer technology that is superior to UTP in many ways.

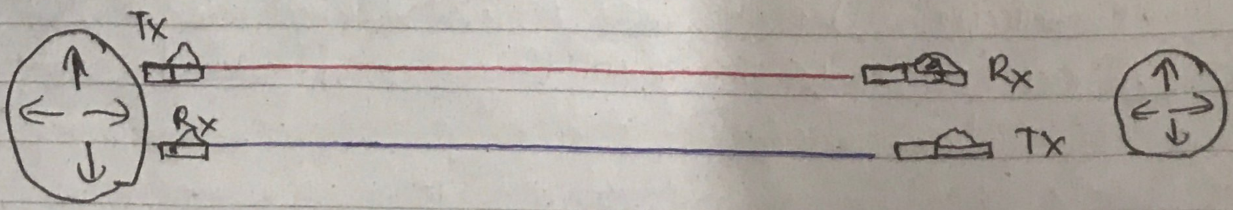
For-eg- copper UTP can be used for upto 100m which is enough for LAN, but what about larger networks. eg Itahari-KTM aba 100m gardai laney kura ta bhyena.

त्यो Router, Switch मा Port हुन्छ (RJ45 को लागि जस्तै but for fiber optics). त्यसमा we plug SFP transceiver. अनि त्यो SFP मा चाहिँ fiber optics को दुइटा a wire jodkam hamiley.

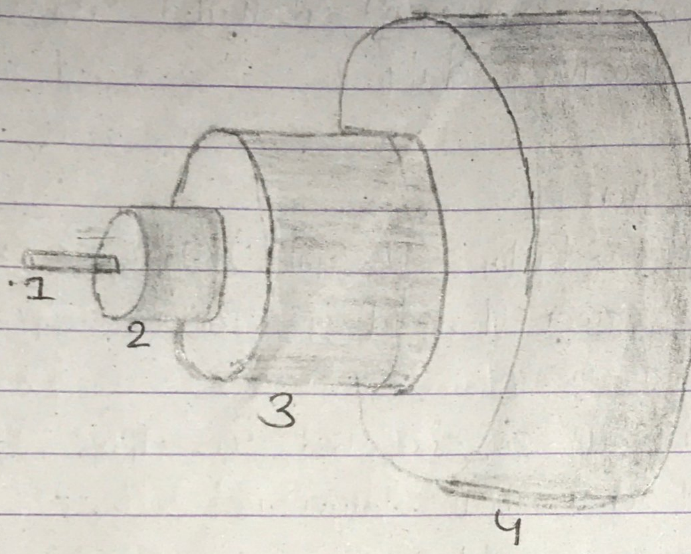
Fiber optics:- They send light over glass fibres.

There are 2 connectors on each end, one to transmit and other to receive. dat on each end.

The copper UTP cables used separate wire pairs within the cable to transmit & receive data. The fibre cable optics instead use separate cables to transmit & receive.



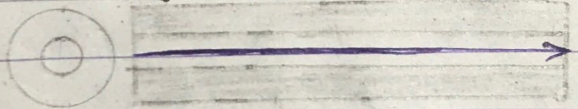
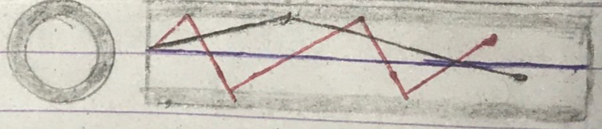
Structure of fibre optics.



1. the fiberglass core itself.
2) cladding that reflects light
3) protective buffer
4) the outerjacket of cable.

There are couple main types of fiber.

i). Single Mode
ii) Multi-Mode

SN	Single Mode fiber	Multimode fiber.
i)		
ii)	Core diameter is narrower than multimode fiber.	• Core diameter is wider than single-mode fiber
iii)	Light enters at a single angle (mode) from a laser based transmitter into fiber-glass core.	• Allows multiple angles (modes) of light waves to enter fiberglass core.
b) •	Allows longer cable than both UTP and multimode fiber	• Allows longer cable than LAN UTP, but shorter cables than single-mode fiber.
•	More expensive than multimode fiber (due to more expensive laser-based SFP transmitter).	• Cheaper than single mode fiber (due to cheaper LED-based SFP transmitters).

Fiber-Optics Cable Standard.

Informal name.	IEEE Standard	Speed	Cable Type	Maximum length.
1000 BASE-LX	802.3z	1Gbps	Multimode or Single-Mode	550m (MM) 5km (SM).
10G BASE-SR	802.3ae	10Gbps	Multimode	400m.
10G BASE-LR.	802.3ae	10Gbps	Single Mode	10km
10G BASE-ER	802.3ae	10Gbps	Single-Mode	30km.

UTP vs Fiber-Optics Cabling.

	UTP	Cabling.
i)	Lower cost that fiber-optic.	Higher cost than UTP.
ii)	Shorter maximum distance than fiber-optic (~100m)	Longer maximum distance than UTP.
iii).	Can be vulnerable to EMI (Electromagnetic Interference) {Strough twist helps}	No vulnerability to EMI.
iv).	RJ45 ports used with UTP are cheaper than SFP ports.	SFP ports are more expensive than RJ45 ports (SM is more expensive than MM).
v)	Emit (leak) a faint signal outside of the cable, which can be copied (= security risk)	Doesn't emit any signal outside of cable (= no security risk).

Quiz

Q)1) You connect two old ~~computers~~ router together with a UTP cable, ~~da~~ however data isn't succesfully sent and received between them. What could be problem?

[✓ a)] Connected with ~~through~~ straight-through cable

b) " " crossover cable

c) " " Auto MDI-X mode

Q.2) Your company wants to connect switches in two separate buildings that are about 150m apart. They want to keep costs down, if possible. What kind of cable should they use.
a) UTP
b) SM - fiber
[✓ c) MM - fiber]

Q.3) Your company wants to connect two offices that are about 3km apart. They want to keep costs down if possible. Which kind of cable should they use?
a). UTP
[✓ b) SM fiber]
c) MM-fiber

Q.4) A switch has following indication over its network interfaces:
10/100/1000Base-T Ports (1-24) -Ports are Auto MDIX.
What would happen if you connect it to an identical switch with a straight-through cable?
[✓ a) They would operate normally.]
b) They would operate at reduced speed
c) they would be unable to communicate.

NOTE.

They wouldn't operate at reduced speed as ports are Auto MDI-X enabled. Even if ports weren't auto MDIX enable, they wouldn't operate at reduced speed. They simply wouldn't operate.

Ports are auto MDI-X enable so they can communicated with straight-through cable.

Q5) एउटै Floor मा ~~क~~ होस्ट hosts हरु, त्यसलाई switch मा connect गर्नु पर्‍यो। which wire to use - Ans UTP!